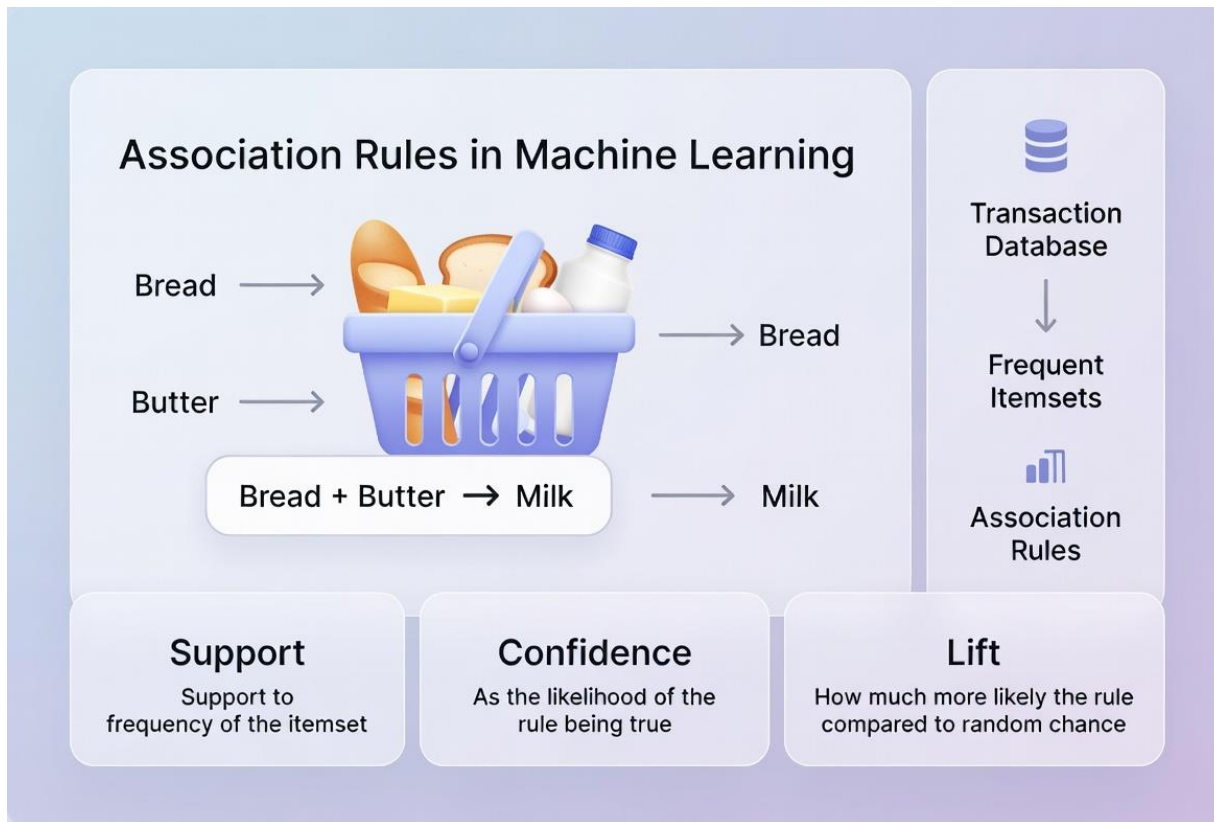


Association Rules

(Association Mining)



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Transactional data

Market basket analysis (MBA) is a data mining technique used to identify relationships between items that are frequently purchased together. The goal is to uncover hidden patterns in transactional data and express them as "**association rules**."



Transactional data

TRANSACTION ID	ITEMS PURCHASED
1	Milk, Bread, Butter
2	Milk, Diapers
3	Bread, Butter, Diapers
4	Milk, Bread, Butter
5	Milk, Bread, Diapers
Total transactions: $N = 5$	

Transactional data is information that is collected and recorded from individual transactions or events. The key characteristic of transactional data is that it captures the details of a specific, single occurrence.

Apriori Algorithm

Apriori Algorithm

Itemset

Support

Confidence

Lift

Apriori Property

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The **Apriori** algorithm, which was first presented by Rakesh Agrawal and Ramakrishnan Srikant in 1994, is used to find frequently occurring itemsets and produce association rules.

Apriori property:

- ✓ If an itemset is frequent, then all of its subsets must also be frequent.
 - If, for example, the itemset {Bread, Butter, Milk} is frequent (i.e., it's bought together often), then it logically follows that the subsets {Bread, Butter}, {Butter, Milk}, {Bread, Milk}, {Bread}, {Butter}, and {Milk} must also be frequent.
- ✓ Contrapositively, if an itemset is infrequent, then all of its supersets must also be infrequent.
 - If the algorithm finds that a small itemset, say {Beer, Chips}, is infrequent, it can immediately "prune" or eliminate all larger itemsets that contain it.

"Frequent" means the itemset appears in a database of transactions more than or equal to a user-defined minimum support threshold.

Support and Confidence

Support: How popular is the itemset?

Support for an itemset X :

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

Confidence: If X , how often is Y also present?

Confidence for a rule $X \rightarrow Y$:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X, Y)}{\text{Support}(X)}$$

Note: An **itemset** is a collection of one or more items. In MBA, it refers to a group of items that appear together in a single transaction.

Lift

Lift: Is the association stronger than random?

Range of Lift: $[0, +\infty)$

- **Lift for a rule $X \rightarrow Y$:**

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)} = \frac{\text{Support}(X, Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

- **Interpretation of Lift:**
 - **Lift > 1:** The occurrence of X and Y together is more likely than would be expected if they were independent. This indicates a strong association.
 - **Lift = 1:** The items X and Y are independent. There is no association between them.
 - **Lift < 1:** The items X and Y are negatively associated. The presence of one reduces the likelihood of the other.

Or these quick formulas

A diagram with the text "Rule $X \Rightarrow Y$ " on the left. Three blue arrows branch out from the right side of this text to the right, pointing to three separate mathematical formulas arranged vertically.

$Support = \frac{Frequency(X,Y)}{N}$

$Confidence = \frac{Frequency(X,Y)}{Frequency(X)}$

$Lift = \frac{Support}{Support(X)*Support(Y)}$

Example dataset

Transaction ID	Items Purchased
T1	Milk, Bread, Butter
T2	Milk, Bread, Eggs
T3	Milk, Eggs

Our goal is to find groups of items that are bought together frequently. Let's set a **minimum support of 50%**. This means an item or itemset must appear in at least 2 out of our 3 transactions.

Step 1: Find Frequent Itemset

a. Count individual items (1-itemsets):

- **{Milk}**: Appears in T1, T2, T3. **Support = $3/3 = 100\%$** (Frequent)
- **{Bread}**: Appears in T1, T2. **Support = $2/3 = 67\%$** (Frequent)
- **{Butter}**: Appears in T1. **Support = $1/3 = 33\%$** (Infrequent, so we discard it)
- **{Eggs}**: Appears in T2, T3. **Support = $2/3 = 67\%$** (Frequent)

Our frequent 1-itemsets are: **{Milk}, {Bread}, {Eggs}**.

b. Count pairs of items (2-itemsets):

Now we look at pairs of the frequent items from the previous step.

- **{Milk, Bread}**: Appears in T1, T2. **Support = $2/3 = 67\%$** (Frequent)
- **{Milk, Eggs}**: Appears in T2, T3. **Support = $2/3 = 67\%$** (Frequent)
- **{Bread, Eggs}**: Appears in T2. **Support = $1/3 = 33\%$** (Infrequent, we discard it)

Our frequent 2-itemsets are: **{Milk, Bread}**, **{Milk, Eggs}**.

Now we need to stop because you cannot have any 3-itemset with a support of at least 50%. Note {milk, bread, eggs} appears in just T2 or just $1/3 = 33.33\%$ support.

Step 2: Generate Association Rules

Now that we have our frequent itemsets, we can create "if-then" rules. Let's set a **minimum confidence of 75%**. This means the rule must be correct at least 75% of the time.

From the frequent itemset {Milk, Bread}:

We can create two possible rules:

- **Rule 1: {Milk} -> {Bread}**
 - **Confidence = Support({Milk, Bread}) / Support({Milk})**
 - Confidence = $(2/3) / (3/3) = 2/3 = 67\%$
 - Since 67% is less than our 75% minimum confidence, this rule is considered **weak and is discarded**.
- **Rule 2: {Bread} -> {Milk}**
 - **Confidence = Support({Milk, Bread}) / Support({Bread})**
 - Confidence = $(2/3) / (2/3) = 100\%$
 - Since 100% is greater than 75%, this is a **strong rule**. It tells us that whenever someone buys bread, they also buy milk.

From the frequent itemset {Milk, Eggs}:

We can create two possible rules:

- **Rule 3: {Milk} -> {Eggs}**
 - **Confidence = $\text{Support}(\{\text{Milk, Eggs}\}) / \text{Support}(\{\text{Milk}\})$**
 - Confidence = $(2/3) / (3/3) = 2/3 = 67\%$
 - This rule is also **weak** and is discarded.
- **Rule 4: {Eggs} -> {Milk}**
 - **Confidence = $\text{Support}(\{\text{Milk, Eggs}\}) / \text{Support}(\{\text{Eggs}\})$**
 - Confidence = $(2/3) / (2/3) = 100\%$
 - This is another **strong rule**. It tells us that whenever someone buys eggs, they also buy milk.

The Final Strong Rules:

Based on our criteria, we have discovered two valuable association rules:

1. **{Bread} -> {Milk}** (Confidence = 100%)
2. **{Eggs} -> {Milk}** (Confidence = 100%)

This information is useful for a store owner. For example, they might place milk near the bread and eggs sections to encourage impulse purchases.

How to use a Lift value

For the rule **{Eggs} -> {Milk}**:

- **Lift = Confidence({Eggs} -> {Milk}) / Support({Milk})**
 - **Lift = 100% / 100% = 1**
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- A lift of exactly 1 means there is **no association** between buying eggs and buying milk.
 - The fact that customers buy milk when they buy eggs is simply because they buy milk in every transaction regardless.
 - If the support for milk was lower (e.g., 50%), a lift greater than 1 would have indicated a stronger, non-coincidental relationship.
 - This demonstrates why lift is an important metric for discovering truly interesting patterns.

For the rule **{Bread} → {Milk}**

$$\text{Lift} = \text{Confidence}(\{\text{Bread}\} \rightarrow \{\text{Milk}\}) / \text{Support}(\{\text{Milk}\}) = 100\% / 100\% = 1$$

How to handle 3-itemsets

Note: We absolutely can stop at 2-itemsets when generating association rules, even if larger itemsets (like 3-itemsets) exist. So, stopping at 2-itemsets is perfectly valid — you're just choosing to extract only pairwise relationships. However, if you need more complex bundles, go ahead with the 3-itemsets.

- $A \wedge B \rightarrow C$
 - $A \wedge C \rightarrow B$
 - $B \rightarrow A \wedge C$
- etc.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X, Y)}{\text{Support}(X)}$$



For a rule like **$A, B \rightarrow C$** , this becomes:

$$\text{Confidence}(A, B \rightarrow C) = \frac{\text{Support}(A, B, C)}{\text{Support}(A, B)}$$

Exercise

Using the earlier transaction dataset:

TRANSACTION ID	ITEMS PURCHASED
1	Milk, Bread, Butter
2	Milk, Diapers
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5	Milk, Bread, Diapers

Total transactions: $N = 5$

Find association rules.

Possible rules from 3-itemsets

A general association rule can be written as:

$$A \rightarrow C$$

Where:

- A is the antecedent (a set of one or more items).
- C is the consequent (a set of one or more items).
- A and C are disjoint, meaning they have no items in common.

Note: All rules must have the union of antecedent and consequent of exactly a 3-itemset, and both sides are non-empty. When generating rules from a specific frequent itemset, all items in that itemset must appear in the rule — either on the left or the right, and none is missing.

If the frequent itemset is **{Milk, Bread, Butter}**, there are six possible association rules that can be generated.

1. Rules with a single item in the consequent:

- Rule 1: {Milk, Bread} $\rightarrow$ {Butter}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Milk, Bread})}$
- Rule 2: {Milk, Butter} $\rightarrow$ {Bread}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Milk, Butter})}$
- Rule 3: {Bread, Butter} $\rightarrow$ {Milk}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Bread, Butter})}$

2. Rules with two items in the consequent:

- Rule 4: {Milk} $\rightarrow$ {Bread, Butter}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Milk})}$
- Rule 5: {Bread} $\rightarrow$ {Milk, Butter}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Bread})}$
- Rule 6: {Butter} $\rightarrow$ {Milk, Bread}
 - Confidence = $\frac{\text{Support}(\text{Milk, Bread, Butter})}{\text{Support}(\text{Butter})}$

So for this 3-itemset, we cannot have just {Milk} $\rightarrow$ {Bread} because the Butter is missing.